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Assessment of groundwater quality for drinking and irrigation purposes: a case study of Peddavanka watershed, Anantapur District, Andhra Pradesh, India

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Abstract In India, the quantity and quality of water available for irrigation is variable from place to place. Assessment of water quality has been carried out to determine the sources of dissolved ions in groundwater. Quality of groundwater in a 398 km² Peddavanka watershed of a semi-arid region of south India is evaluated for its suitability for drinking and irrigation purposes. The middle Proterozoic Cuddapah Supergroup and Kurnool Group of rocks underlie most of the watershed. The main lithologic units consist chiefly of quartzite, limestone, and shale. Seventy-six water samples were collected from open-wells and bore-holes. Water samples were collected representative of the post-monsoon (winter) and pre-monsoon (summer). The quality assessment is made through the

estimation of Ca²⁺, Mg²⁺, Na⁺, K⁺, Cl⁻, SO₄²⁻, CO₃²⁻, HCO₃⁻, total hardness as CaCO₃, TDS, EC, and pH. Based on these analyses, parameters like sodium adsorption ratio, % sodium, residual sodium carbonate, non-carbonate hardness, potential salinity, Kelley's ratio, magnesium ratio, index of base exchange and permeability index were calculated. According to Gibbs' ratio samples in both seasons fall in the rock dominance field. The overall quality of waters in the study area in post-monsoon season is high for all constituents ruling out pollution from extraneous sources.

Keywords Groundwater quality assessment · Drinking · Irrigation · Peddavanka watershed · Andhra Pradesh · India

Introduction

Agriculture is a dominant sector in the economic development of India, as it is the source of sustenance for the majority of the population, and contributes 46% of the gross national product (Singh 1983). The Peddavanka river rises from Mutchukota hills of the Er-ramalai hills and flows in a southern direction to join the Pennar river near Tadipatri of the Anantapur district, Andhra Pradesh, India. Peddavanka watershed covers an area of 398 Km², between Latitudes 14°55'24"–15°12'50" N and Longitudes 77°51'20" – 78°03'26" E

(Fig. 1). The climate of the watershed is hot and semi-arid and receives an average annual rainfall of 550 mm, out of which 90 % is received during the southwest (June-Sept) and the northeast (Oct-Dec) monsoon periods. The chemical quality of groundwater is related to the lithology of the area and the residence time the water in contact with rock material. Weathered mantle, soils, and, atmosphere are the important factors responsible for contribution of dissolved solids to water. It is impossible to control the dissolution of undesirable constituents in the waters after they enter the ground (Johnson 1979; Sastri 1994).

Geology

The watershed is primarily underlain by the Tadipatri formation of the middle Proterozoic Cuddapah Super group, which is predominantly by shale with inter-bedded layers of flaggy limestone with quartzite as an outlier. The shale trends NNE-SSW with an $\sim 10^\circ$ dip towards the east. Shale contains layers of basic volcanic rocks, mainly sills of trap and trap conglomerate with volcanic pellets. The Tadipatri formation is overlain by the rocks of the Kurnool group, which include the Banaganapalle quartzite, Narji limestone and Auk shale. A distinct angular unconformity separates the Cuddapah formation from the Kurnool rocks, the latter occurring as horizontal beds over the tilted Cuddapah rocks (Ramam and Murty 1997; Gowd SS 1998). Alluvium of recent age comprises sand, silt, and clay, which occur all along the course of the stream and at the confluence of the stream with Pennar River in the southern portion of the watershed (Fig. 2). The geological succession in the watershed is as follows.

Stratigraphic succession (*after* Murthy et al. 1979)

	Series	Stage	Thickness
Local disconformity Kurnool Group	Paniam	Plateau Quartzite	10–35 m
	Jammalamadugu	Auk Shales	10–30 m
		Narji Limestones (massive limestone and Flaggy limestone)	100–200 m
	Banaganapalli	Quartzite with Conglomerate	10–57 m
Unconformity Cuddapah Super Group	Chitravati Group	Tadipatri formation	4600 m
		Shale& basic intrusives: sills	1–75 m

Methodology

Samples were collected from groundwater and surface water representing pre and post-monsoon seasons to evaluate the seasonal variations in chemical composition. Techniques and methods followed for collection, preservation, analyses and interpretation of water samples are those given by standard methods (APHA 1995).

A total of seventy-six water samples over the watershed were collected: four surface water samples S1, S2, S3, and S4 (Fig. 3). The remainder of the water samples were collected from open and bore-wells, twice

from each location during post-monsoon (winter) and pre-monsoon (summer).

Silica content was determined by Molybdate Blue method using a field Elico spectrophotometer. Na^+ and K^+ were measured by using a flame photometer. Total hardness as CaCO_3 , calcium (Ca^{2+}), carbonate (CO_3^{2-}), bicarbonate (HCO_3^-) and, chloride (Cl^-) were analysed by volumetric methods. Mg^{2+} was calculated from total hardness and calcium. Sulphates (SO_4) were measured using the colorimetric technique. Total dissolved solids (TDS) were measured by evaporation and calculation methods (Hem 1991). EC and pH of water samples were measured in the field immediately after the collection of samples using a portable field kit. Except pH, which is expressed as dimensionless number, all other constituents are expressed in mg/l. Non-Carbonate hardness was calculated using the values of hardness and alkalinity. Iron, manganese and fluoride were found only in traces and hence not included.

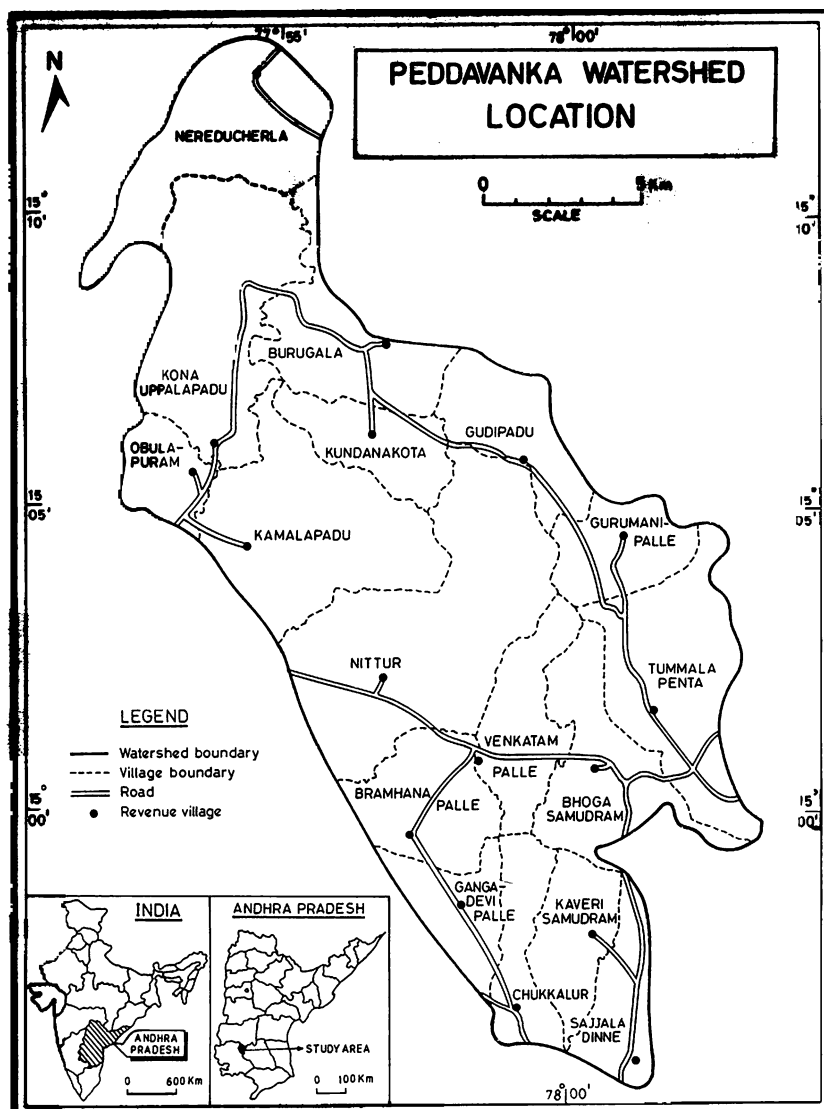
Results and discussion

The analytical results, computed values and the statistical parameters like minimum, maximum, mean and the standard deviation values of water samples are given in Tables 1 and 2. Elemental concentration and its seasonal variations in post and pre-monsoon are given in Figures 4 and 5.

The total dissolved solids (TDS) measured by residue on evaporation method range from 301 to 1487 mg/l with a mean of 638 mg/l in post-monsoon season; the same in the pre-monsoon water samples range from 256 to 1226 mg/l with a mean of 530 mg/l. Based on the classification of TDS (WHO 1989; USEPA 1992), the waters of the basin in both the seasons are useful for irrigation, while 90% are useful in post-monsoon and 94% are useful in pre-monsoon seasons for drinking.

Specific conductance varies from 510 to 2520 micromhos/cm at 25°C with a mean of 1075 micromhos/cm at 25°C for post-monsoon water samples while it varies from 365 to 1830 micromhos/cm at 25°C for pre-monsoon samples. Dissolved solids of water can be indirectly estimated using specific conductance (Hem, 1991). The multiplication factor computed to estimate the basin waters varies from 0.57 TO 0.66 with an average of 0.59 in post-monsoon samples and from 0.60 to 0.74 with an average of 0.68 in pre-monsoon samples, which are very much within the limits as recommended by Hem (1991). The hydrogen-ion-concentration (pH) in post-monsoon samples varies from 6.8 to 8.8 with a mean of 8.36 while in the pre-monsoon samples it varies

Fig. 1 Location map of Peddavanka watershed



from 6.7 to 8.6 with a mean of 8.45 indicating their alkaline nature.

Agronomic aspects

The major parameters have been grouped in two categories namely Salinity and Sodicity (Rhodes 1972; Ayers and Westcot 1985; WHO 1989; Kuchanwar et al., 1999), which may affect the soil, plant and human use directly or indirectly.

Sodium Adsorption Ratio

Irrigation waters are classified based on sodium adsorption ratio (WHO 1989). Based on the classifica-

tion, 97 % of the water samples of the watershed area fall in excellent category and 3 % samples falls in good category in post-monsoon season and 99 % of the samples falls in the excellent category and 1 % falls in the good category in pre-monsoon season. Therefore, none of the samples are of the poor category for irrigation in either season. A more detailed analysis, however with respect to the irrigation suitability of the water was made by plotting the data on US Salinity Laboratory diagram (Figs 6a and 6b).

Percent Sodium

Wilcox (1955) used % sodium and specific conductance in evaluating irrigation waters using Wilcox diagram as given in Figure 7a and 7b.

The Wilcox diagram is divided into five areas and the waters are rated as given below.

Season	Area	No. Of water samples	Percentage
Post-monsoon	Excellent to Good	26	34
	Good to Permissible	33	43
	Permissible to Doubtful	6	8
	Doubtful to Unsuitable	11	15
Pre-monsoon	Excellent to Good	40	58
	Good to Permissible	15	22
	Permissible to Doubtful	14	20

Most of the water samples in two seasons fall in excellent to good and good to permissible areas indicating their usefulness for irrigation. No water sample is strictly unsuitable for irrigation.

Residual Sodium Carbonate (RSC)

The following table gives the results and rating of waters based on residual sodium carbonate (WHO 1989).

RSC meq / l	Irrigation Class	Number of samples			
		Post-monsoon		Pre-monsoon	
		No.	%	No.	%
< 1.25	Safe	45	59	38	55
1.25–2.50	Marginal	12	16	13	19
> 2.50	Unsuitable	19	25	18	26

Fig. 2 Geological map of Peddavanka watershed

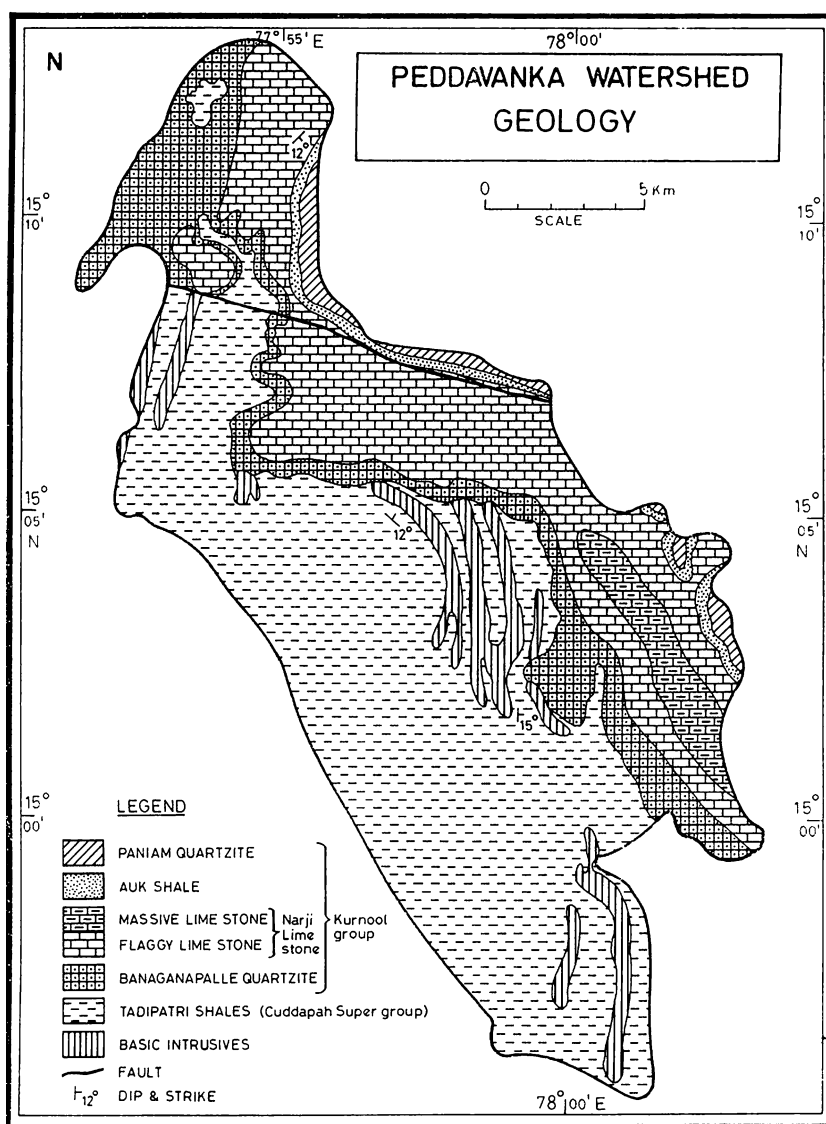
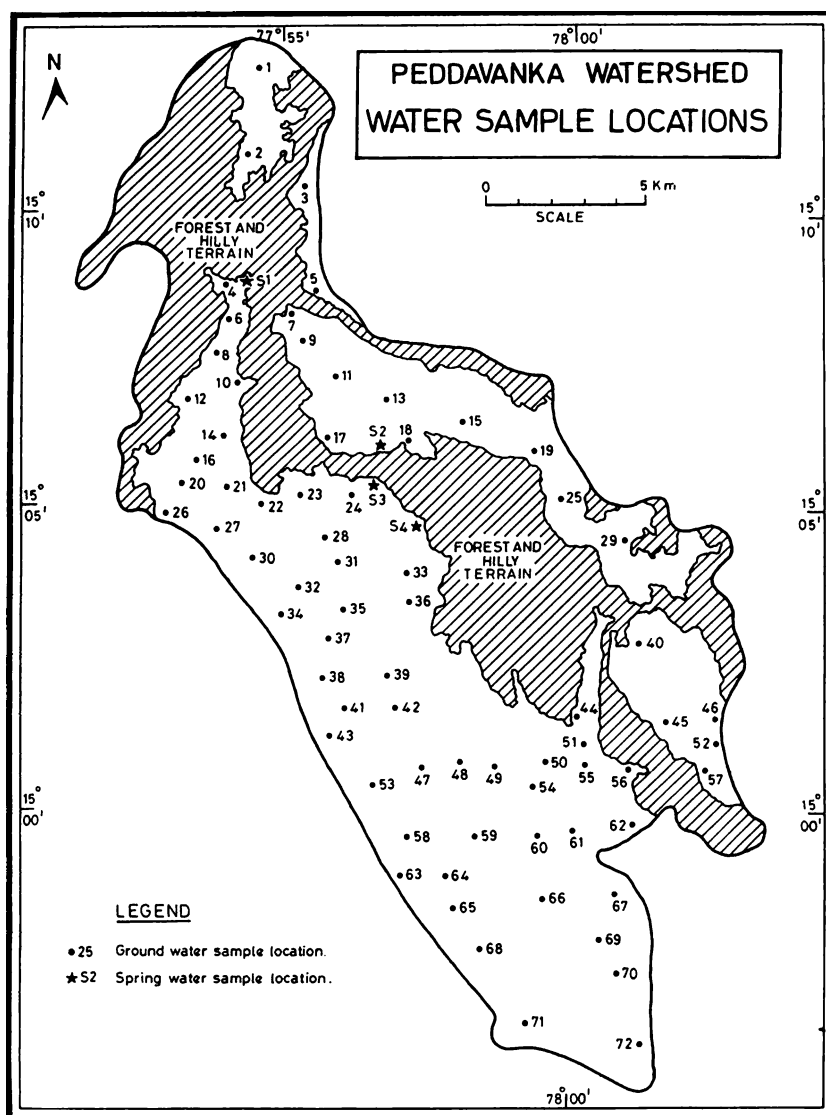


Fig. 3 Map depicting the water sample locations



The residual sodium carbonate varies from 0 to 17.56 with an average of 3.36 in post-monsoon waters while it varies from 0 to 7.86 with an average of 2.70 in pre-monsoon waters. The water quality declines in post-monsoon season, however the unsuitable category is about 25%. Poor agricultural returns are partly owing to this reason. The residual sodium carbonate distribution and the iso-concentration diagrams are depicted in Figures 8a and 8b.

Kelley's Ratio

Sodium measured against Ca^{2+} and Mg^{2+} was considered by Kelley (1951) to calculate the Kelley's ratio and it varies from 0.05 to 15.99 in the post-monsoon samples while it ranges from 0.01 to 5.70 in those of the pre-monsoon waters. Therefore according to Kelley's ratio,

68% of the samples in the post-monsoon period are suitable for irrigation and 64% for the pre-monsoon waters (Figures 9a and 9b).

Magnesium Ratio

Generally calcium and magnesium maintain a state of equilibrium in most waters. More Mg^{2+} present in waters will adversely affect the soil quality converting it to alkaline and decreases crop yields. Rocks in the watershed contain more magnesium (dolomite and basalt), and most waters contain Mg^{2+} more than Ca^{2+} . Seventy-six percent of the post-monsoon waters contain magnesium ratio greater than 50% while 78% of pre-monsoon waters contain magnesium ratio greater than 50%.

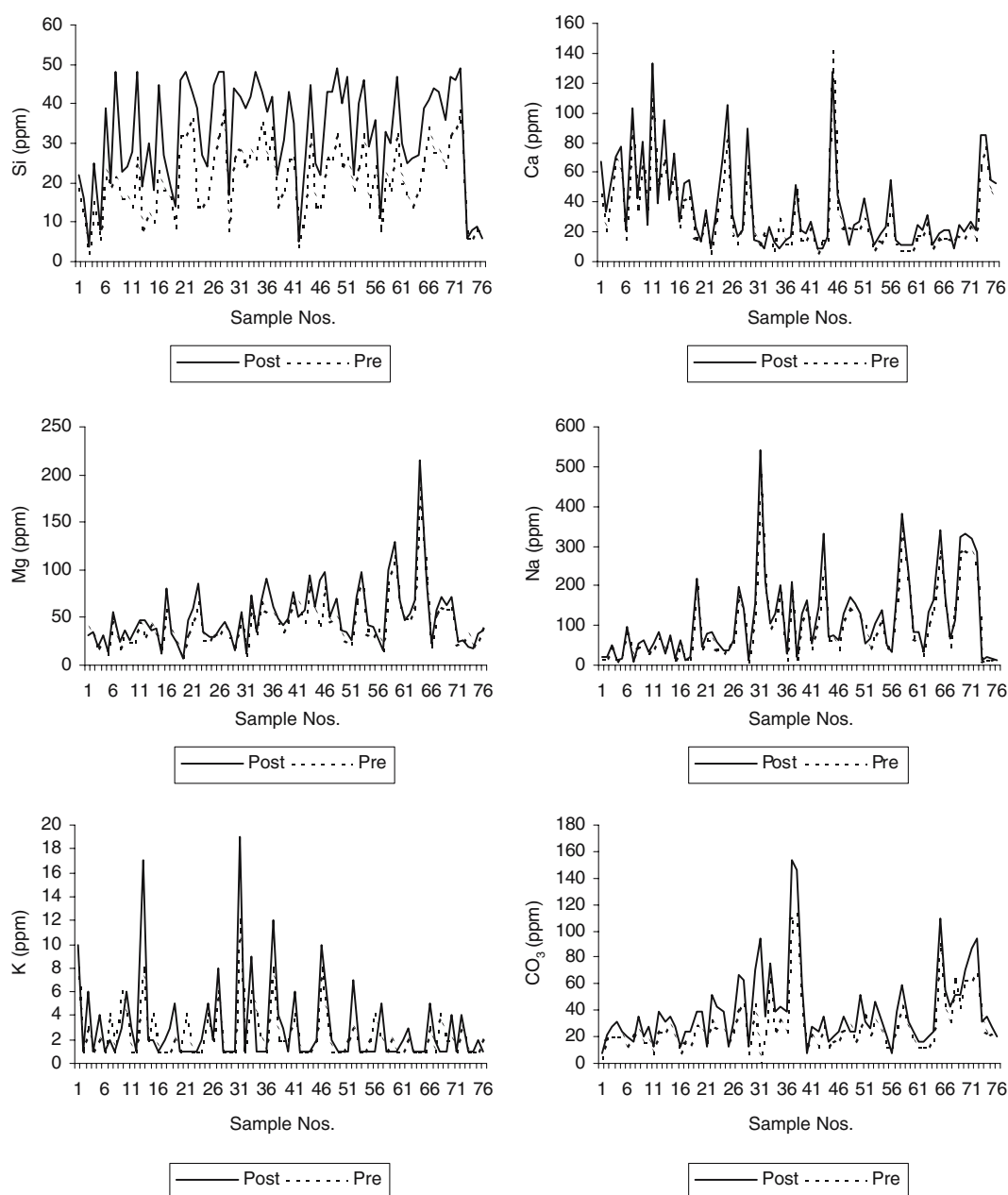


Fig. 4 Elemental concentrations and its variations in post-monsoon and pre-monsoon seasons for Si, Ca, Mg, Na, K, and CO₃

Gibbs' ratio

Gibbs' diagram is widely used to establish the relationships of water composition and aquifer lithology (Gibbs 1970) (Figures 10a and 10b). Almost all the samples from post-monsoon and pre-monsoon seasons fall in the rock dominance area indicating the interaction between rock chemistry and the chemistry of the percolating waters in the subsurface.

Index of Base Exchange

It is essential to know the changes in chemical composition of groundwater during its travel in the sub-surface (Sastri 1994). The Chloro-alkaline indices CAI 1 and CAI 2 are suggested by Schoeller (1977), which indicate the ion exchange between the groundwater and its host environment. The Chloro-alkaline indices used in the evaluation of Base Exchange are calculated using the formulae.

1. Chloro Alkaline Indices 1 = $\frac{Cl - (Na + K)}{Cl}$
2. Chloro Alkaline Indices 2 = $\frac{Cl - (Na + K)}{(SO_4 + HCO_3 + CO_3 + NO_3)}$

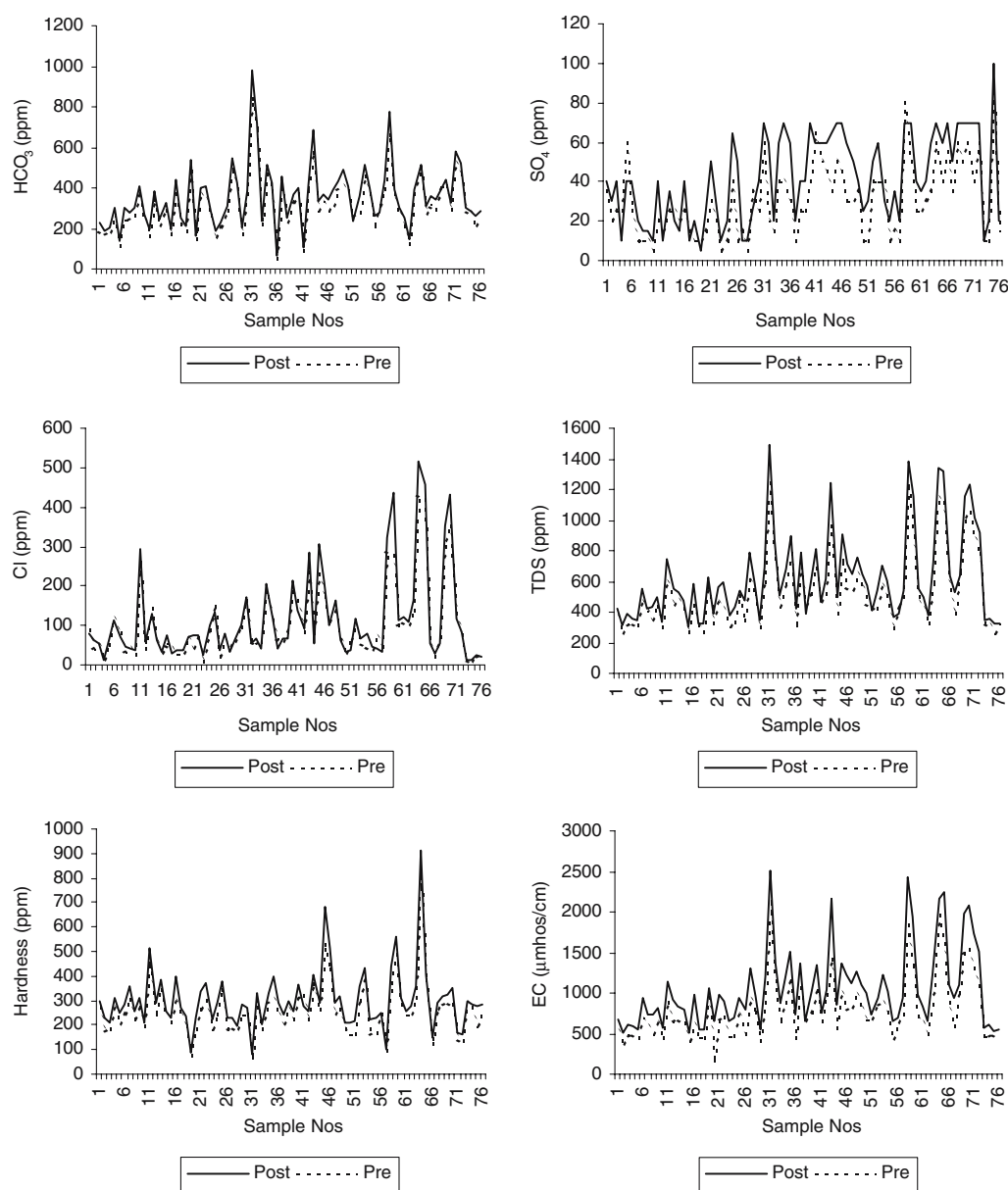


Fig. 5 Elemental concentrations and its variations in post-monsoon and pre-monsoon seasons for HCO₃, SO₄, Cl, TDS, EC, and Hardness

If there is ion exchange of Na⁺ and K⁺ from water with magnesium and calcium in the rock, the exchange is known as direct when the indices are positive. If the exchange is reverse then the exchange is indirect and the indices are found to be negative.

The Chloroalkaline Indices are calculated for the post-monsoon and pre-monsoon waters of the watershed as given in Tables 1 and 2. Noted that 72% of the post-monsoon and 71% of the pre-monsoon waters show negative ratios and 28% of the post-monsoon and

29% of the pre-monsoon waters show positive ratios. Seasonally there is either deviation in the Chloroalkaline Indices.

Permeability Index

The soil permeability is affected by long-term use of irrigation water as influenced by Na⁺, Ca²⁺, Mg²⁺, and HCO₃⁻ contents of the soil. WHO (1989) uses a criterion for assessing the suitability of water for irrigation based on permeability index. According to the permeability index values, 28% of the groundwater comes under class I (P.I. > 7.5%) category and 72% comes under class II (P.I. ranges from 25 to 75%) category in post-monsoon,

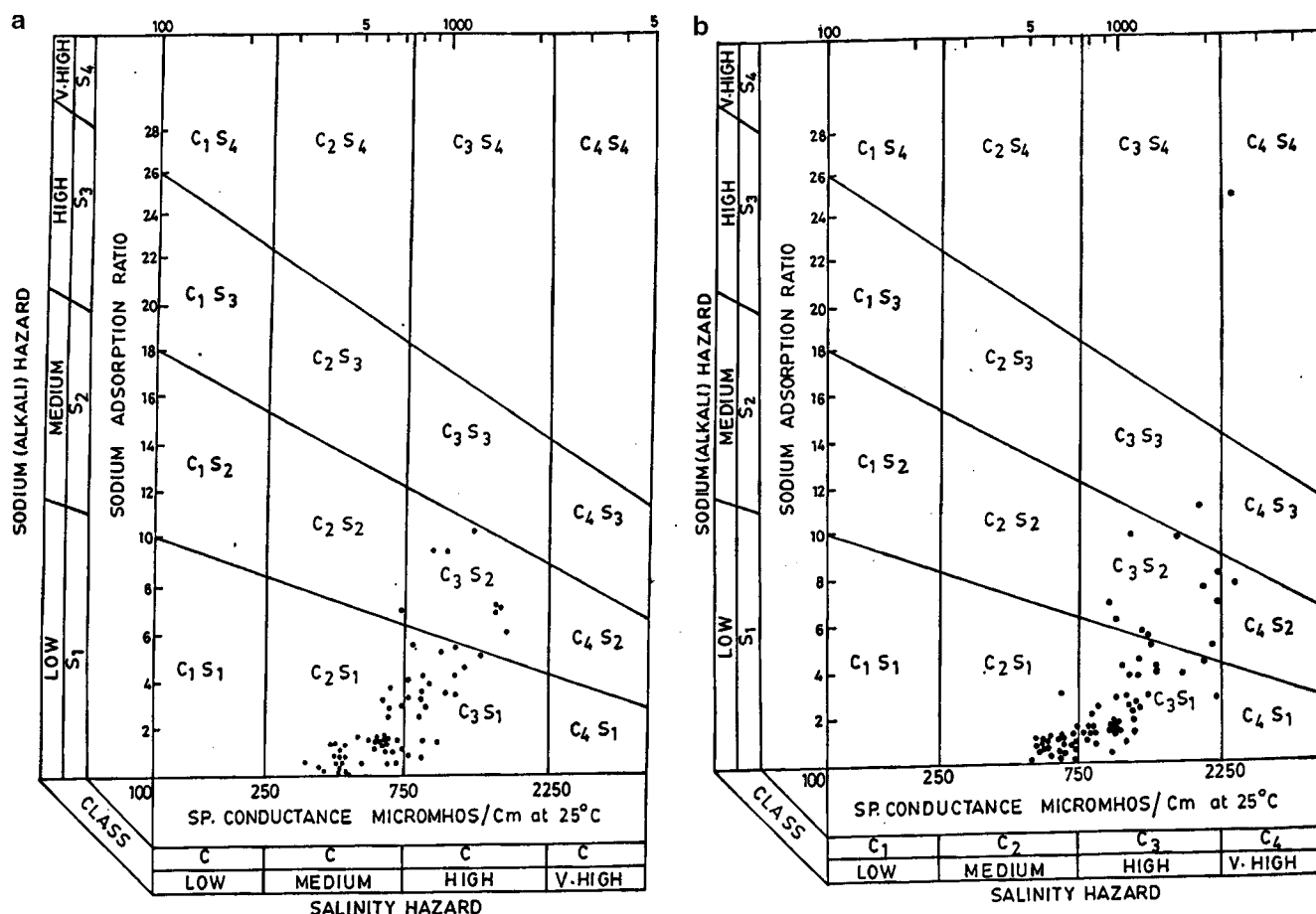


Fig. 6 Rating of water samples in relation to salinity and sodium hazard (USSL 1954) a) Pre-monsoon b) Post-monsoon

while 32% comes under class I (P.I. > 7.5%) category and 68% comes under class II (P.I. ranges from 25 to 75%) category in pre-monsoon season.

Conclusions

Groundwater in the region is generally alkaline and shows marginal seasonal changes. The hydrogen-ion-concentration in post-monsoon samples varies from 6.8 to 8.8 and in the pre-monsoon samples it varies from 6.7 to 8.6 indicating their alkaline nature. Based on the concentration of TDS, the waters of the basin in the seasons are useful for irrigation, while 90% are useful in post-monsoon and 94% are useful in pre-monsoon seasons for drinking. Based on Wilcox 1955; WHO 1989 classification, 67% of the waters of the post-monsoon belong to excellent to good and good to permissible category while about 80% of the pre-monsoon belong to the same categories mentioned above.

Irrigation waters classified based on SAR alone has indicated that out of 76 samples, 97% of the water samples belong to excellent category while others are of good category in both seasons. Graphical representation of the chemical data on the irrigation suitability diagram (USSL, 1954) shows that medium salinity – low sodium (C2 S1) and high salinity-low sodium (C3 S1) waters are present which need adequate drainage to overcome the salinity problem. According to WHO (1989) classification of water based on sodium adsorption ration (SAR), 97 % samples are good category for irrigation. According to Kelley's ratio 64 % of the samples in the pre-monsoon period and 68 % of the samples in the post-monsoon period are suitable for irrigation. As the rocks in the watershed area contain more Mg^{2+} bearing minerals it is observed that most waters contain Mg^{2+} more than Ca^{2+} .

Based on Gibbs' ratios of water samples plotted against total dissolved solids, water samples from pre and post-monsoon seasons fall in the rock dominance area indicating the interaction between rock chemistry and the chemistry of the percolating precipitation waters in the sub-surface. The chloroalkaline indices for post and pre-monsoon water of 28% belong to the positive ratios while 72% of the respective seasons belong to the

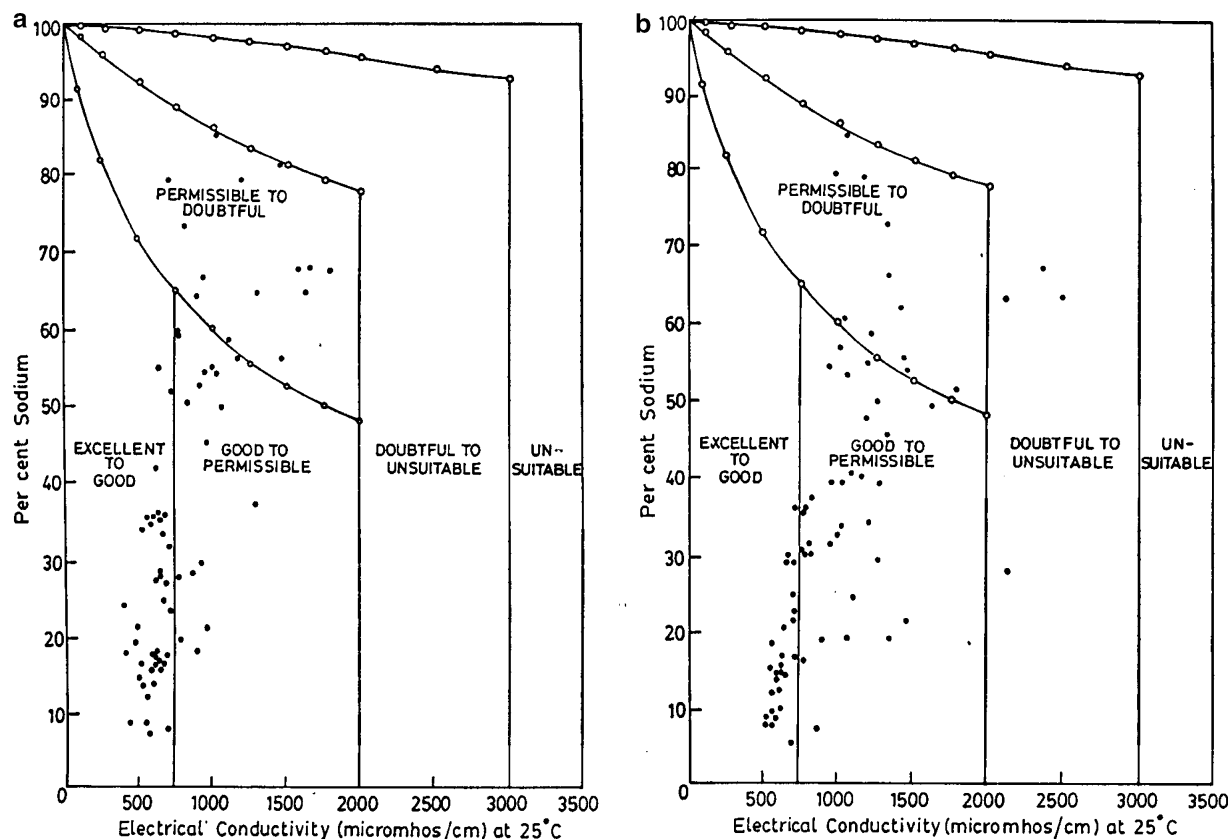


Fig. 7 Specific conductance and % sodium relationship for rating irrigation water (Wilcox 1955) a) Pre-monsoon b) Post-monsoon

negative index indicating that the host rocks are primary sources of dissolved solids in the water. According to the permeability index values, pre and post-monsoon waters

comes under class II category, where permeability index ranges from 25–75%.

The quality of groundwater is controlled by lithology apart from other factors like land use pattern and palaeoenvironmental conditions. In general, the TDS increases after rains as a result of dissolution of minerals

Fig. 8 Residual sodium carbonate (RSC) distribution map a) Pre-monsoon b) Post-monsoon

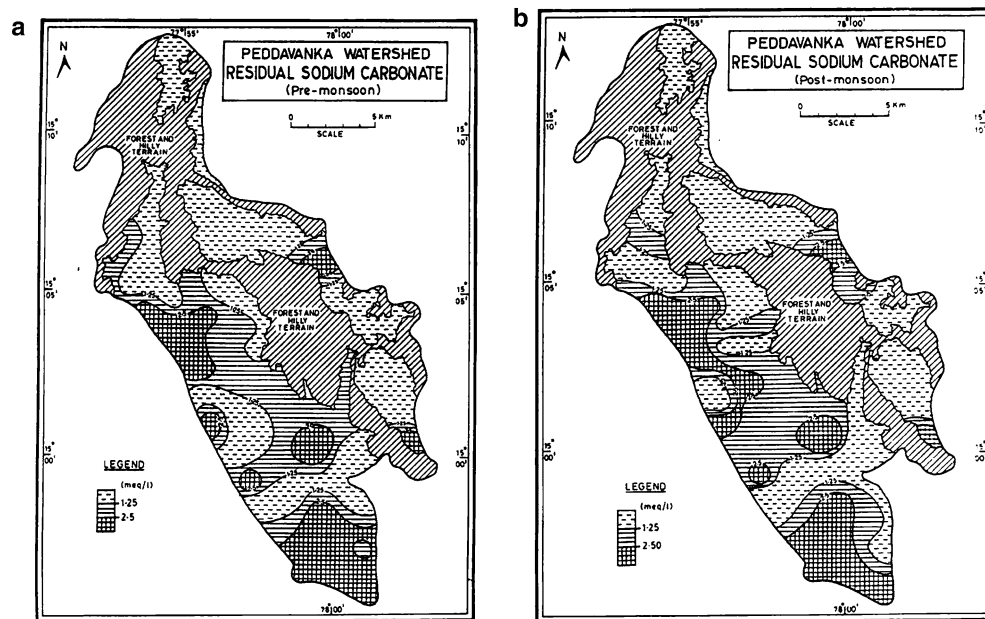


Fig. 9 Distribution of Kelley's ratio a) Pre-monsoon b) Post-monsoon

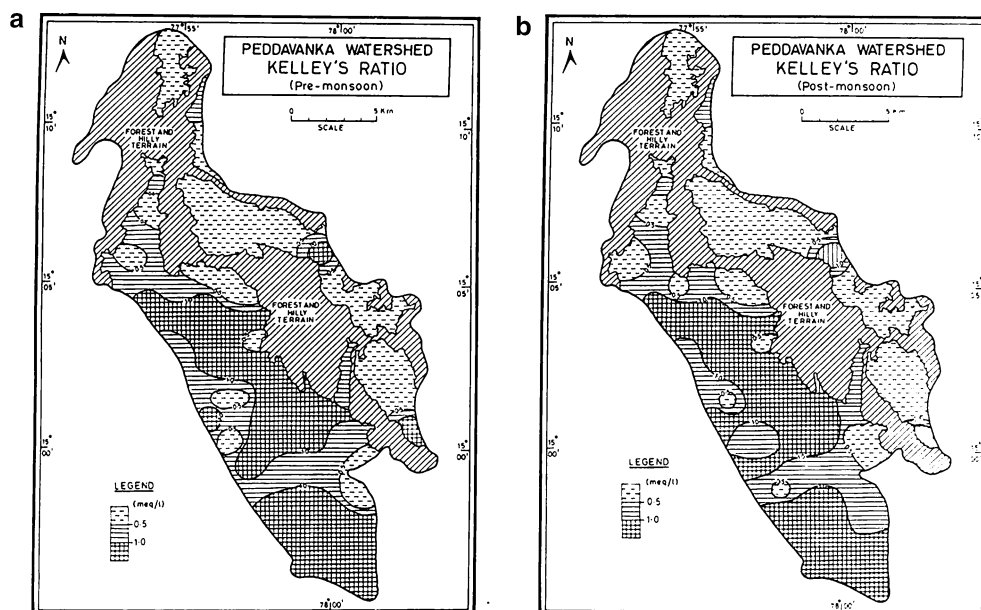


Table 1 Statistical parameters of groundwater samples (Post-monsoon)

S.No	Constituents	Units	Min.	Max.	Mean	Standard deviation
1	Silica	Mg/l	4	49	33.67	12.03
2	Calcium	Mg/l	9	133	35.31	28.78
3	Magnesium	Mg/l	7	215	52.06	31.73
4	Sodium	Mg/l	9	540	119.92	104.41
5	Potassium	Mg/l	1	19	3.11	3.58
6	Carbonate	Mg/l	0	153	38.59	26.02
7	Bicarbonate	Mg/l	64	772	359.28	154.60
8	Sulphate	Mg/l	5	70	41.94	21.03
9	Chloride	Mg/l	11	514	116.93	111.49
10	Total dissolved solids	Mg/l	301	1487	637.69	283.43
11	Hardness as CaCO ₃	Mg/l	80	912	302.29	124.58
12	Alkalinity as CaCO ₃	Mg/l	85	882	324.88	137.38
13	Non-carbonate hardness	Mg/l	0	469	108.97	108.23
14	Specific conductance	$\mu S / cm$ at 25° C	510	2520	1074.78	478.86
15	PH		6.8	8.8	8.36	0.38
16	Sodium adsorption ratio		0.09	25.97	3.30	3.71
17	Percent sodium		6	94	41.07	20.84
18	Potential salinity		0.41	15.23	3.76	3.29
19	Residual sodium carbonate		0	17.56	3.39	3.35
20	Gibbs Ratio I		0.05	0.70	0.28	0.16
21	Gibbs Ratio II		0.11	0.97	0.65	0.28
22	Magnesium Ratio		21.32	96.98	67.06	21.85
23	Kelley's Ratio		0.05	14.36	1.10	1.85

from the overlying material by the infiltrating waters. The overall quality of waters in the watershed in post-monsoon season where all the constituents are high because of rock-water interaction ruling out any pollution

from extraneous sources. According to the overall assessment of the watershed, water quality was found to be useful for drinking and irrigation purposes, barring a few patches in the west and southwest.

Table 2 Statistical parameters of groundwater samples (Pre-monsoon)

S.No	Constituents	Units	Min.	Max.	Mean	Standard deviation
1	Silica	Mg/l	2	38	21.76	8.48
2	Calcium	Mg/l	8	141	31.24	25.49
3	Magnesium	Mg/l	8	113	43.00	20.78
4	Sodium	Mg/l	9	342	100.48	81.00
5	Potassium	Mg/l	1	8	2.44	1.96
6	Carbonate	Mg/l	4	110	28.52	19.47
7	Bicarbonate	Mg/l	48	684	310.30	125.84
8	Sulphate	Mg/l	5	80	29.24	16.88
9	Chloride	Mg/l	10	418	101.82	86.00
10	Total dissolved solids	Mg/l	256	1226	529.58	214.77
11	Hardness as CaCO ₃	Mg/l	72	524	254.65	86.19
12	Alkalinity as CaCO ₃	Mg/l	59	581	276.38	110.56
13	Non-carbonate hardness	Mg/l	0	310	102.68	76.69
14	Specific conductance	$\mu S / cm$ at 25° C	365	1830	778.02	318.33
15	pH		6.7	8.6	8.45	0.34
16	Sodium adsorption ratio		0.24	10.25	2.88	2.52
17	Percent sodium		7	85	41.38	20.44
18	Potential salinity		0.33	12.21	3.18	2.53
19	Residual sodium carbonate		0	7.86	2.70	2.14
20	Gibbs Ratio I		0	0.71	0.27	0.18
21	Gibbs Ratio II		0	0.97	0.58	0.32
22	Magnesium Ratio		0	85.0	36.34	23.29
23	Kelley's Ratio		0	5.7	0.89	1.07

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